

Applications of Integration

Ex9.1

Question 1.

Find an approximate value of $\int_1^{1.5} x dx$ by applying the left-end rule with the partition $\{1.1, 1.2, 1.3, 1.4, 1.5\}$

Solution:

Here $a=1$, $b=1.5$, $n=5$, $f(x)=x$

So, the width of each subinterval is

$$h=\Delta x=\frac{b-a}{n}=\frac{1.5-1}{5}=0.1 \quad x_0=1; x_1=1.1; x_2=1.2; x_3=1.3; x_4=1.4; x_5=1.5$$

The left hand rule for Riemann sum,

$$S=[f(x_0)+f(x_1))+f(x_2)+f(x_3)+f(x_4)]\Delta x=[f(1)+f(1.1)+f(1.2)+f(1.3)+f(1.4)]$$

$$(0.1)=[1+1.1+1.2+1.3+1.4](0.1)=[6](0.1)=0.6.$$

Question 2.

Find an approximate value of $\int_1^{1.5} x^2 dx$ by applying the right-end rule with the partition $\{1.1, 1.2, 1.3, 1.4, 1.5\}$

Solution:

Here $a=1$;

$$b=1.5, n=5; f(x)=x^2$$

So, the width of each subinterval is

$$h=\Delta x=\frac{b-a}{n}=\frac{1.5-1}{5}=0.1 \quad x_0=1; x_1=1.1; x_2=1.2; x_3=1.3; x_4=1.4; x_5=1.5$$

The Right hand rule for Riemann sum,

$$S=[f(x_1)+f(x_2)+f(x_3)+f(x_4)+f(x_5)]\Delta x=[f(1.1)+f(1.2)+f(1.3)+f(1.4)+f(1.5)]$$

$$(0.1)=[1.21+1.44+1.69+1.96+2.25](0.1)=[8.55](0.1)=0.855.$$

Question 3.

Find an approximate value of $\int_1^{1.5} (2-x) dx$ by applying the mid-point rule with the partition $\{1.1, 1.2, 1.3, 1.4, 1.5\}$

Solution:

Here $a=1$

$$b=1.5, n=5, f(x)=2-x$$

So, the width of each subinterval is

$$h=\Delta x=\frac{b-a}{n}=\frac{1.5-1}{5}=0.1 \quad x_0=1; x_1=1.1; x_2=1.2; x_3=1.3; x_4=1.4; x_5=1.5$$

The mid-point rule for Riemann sum,

$$S=[f(x_0+x_1/2)+f(x_1+x_2/2)+f(x_2+x_3/2)+f(x_3+x_4/2)+f(x_4+x_5/2)]\Delta x=[f(1+1.1/2)+f(1.1+1.2/2)+f(1.2+1.3/2)+f(1.3+1.4/2)+f(1.4+1.5/2)]$$

$$=[f(1.05)+f(1.15)+f(1.25)+f(1.35)+f(1.45)]$$

$$(0.1)=[0.95+0.85+0.75+0.65+0.55](0.1)=[3.75](0.1)=0.375.$$

Ex 9.2

Question 1.

Evaluate the following integrals as the limits of sums:

(i) $\int_0^1 (5x+4) dx$

Solution:

We use the formula

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \Delta x \sum_{r=1}^n f(a+r\Delta x) \quad \text{Here, } a=0; b=1; f(x)=5x+4 \quad \Delta x=\frac{b-a}{n}=\frac{1-0}{n}=\frac{1}{n}$$

$$f(a+r\Delta x)=f(0+r \cdot \frac{1}{n})=f(\frac{r}{n})=5(\frac{r}{n})+4=5\frac{r}{n}+4$$

$$\Rightarrow \int_0^1 (5x+4) dx = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n (5\frac{r}{n}+4) = \lim_{n \rightarrow \infty} \frac{1}{n} [5 \sum_{r=1}^n \frac{r}{n} + \sum_{r=1}^n 4] = \lim_{n \rightarrow \infty} [5 \sum_{r=1}^n \frac{r}{n^2} + 4 \sum_{r=1}^n \frac{1}{n}]$$

$$= \lim_{n \rightarrow \infty} [5 \frac{1}{n^2} \frac{n(n+1)}{2} + 4 \frac{1}{n} n] = \lim_{n \rightarrow \infty} [5 \frac{(1+\frac{1}{n})}{2} + 4]$$

Applying limit we get,

$$=52+4=132.$$

$$(ii) \int_2^1 (4x^2 - 1) dx$$

Solution:

We use the formula

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$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \Delta x \sum_{r=1}^n f(a+r\Delta x) \text{ Here, } a=1; b=2; f(x)=4x^2-1 \Delta x = b-a = 2-1 = 1/n$$

$$f(a+r\Delta x) = f(1+r \cdot 1/n) = f(1+r/n) = 4(1+r/n)^2 - 1 = 4[1 + r^2/n^2 + 2r/n] - 1 = 4 + 4r^2/n^2 + 8r/n - 1 = 4r^2/n^2 + 8r/n + 3 \quad (1)$$

$$\Rightarrow \int_2^1 (4x^2 - 1) dx = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n (4r^2/n^2 + 8r/n + 3) = \lim_{n \rightarrow \infty} [4/n^3 \sum_{r=1}^n r^2 + 8/n^2 \sum_{r=1}^n r + 1/n \sum_{r=1}^n 3] = \lim_{n \rightarrow \infty} [4/n^3 \cdot n(n+1)(2n+1)/6 + 8/n^2 \cdot n(n+1)/2 + 1/n \cdot 3n]$$

$$= \lim_{n \rightarrow \infty} [2/3(1+1/n)(2+1/n) + 4(1+1/n) + 3] = 2/3(1)(2) + 4(1) + 3 = 4/3 + 4 + 3 = 4/3 + 7 = 25/3.$$

Ex 9.3

Question 1.

Evaluate the following definite integrals :

$$\int_4^3 dx x^2 - 4$$

Solution:

Let

$$I = \int_4^3 dx x^2 - 4 = \int_4^3 dx x^2 - 2 \int_4^3 \frac{dx}{x} \quad [\because \int \frac{dx}{x} = \log(x)]$$

$$= [12 \log(x) - 2x^2]_4^3 = [12 \log(3) - 2(9)] - [12 \log(4) - 2(16)]$$

$$= 12 \log(3) - 18 - 12 \log(4) + 32 = 12 \log(3) - 12 \log(4) + 14$$

$$= 12 [\log(3) - \log(4)] + 14 = 12 [\log(3/4)] + 14 = 12 \log(3/4) + 14$$

$$(ii) \int_1^{-1} dx x^2 + 2x + 5$$

Solution:

$$\text{Let } I = \int_1^{-1} dx x^2 + 2x + 5$$

$$= \int_1^{-1} dx (x+1)^2 - 1 + 5 = \int_1^{-1} dx (x+1)^2 + (2)^2 \int_1^{-1} \frac{dx}{x+1} = [12 \tan^{-1}(x+1)]_1^{-1} - 12 [\tan^{-1}(2) - \tan^{-1}(0)]$$

$$= 12 [\tan^{-1}(2) - \tan^{-1}(0)] - 12 [\tan^{-1}(1) - \tan^{-1}(0)] = 12 [\pi/4 - 0] = \pi/8.$$

$$(iii) \int_0^1 \sqrt{1-x+x^2} dx$$

Solution:

$$\text{Let } I = \int_0^1 \sqrt{1-x+x^2} dx$$

$$\text{Put } x = \cos 2\theta$$

x	0	1
θ	$\frac{\pi}{4}$	0

$$\text{Let } I = \int_0^1 \sqrt{1-x+x^2} dx$$

$$\text{Put } x = \cos 2\theta$$

x	0	1
θ	$\frac{\pi}{4}$	0

Differentiate with respect to θ

$$dx = -2 \sin 2\theta d\theta$$

Consider:

$$\sqrt{1 - \cos 2\theta} = \sqrt{2 \sin^2 \theta} = \sin \theta \quad (\sin 2\theta) = 2 \sin \theta \cos \theta$$

$$2(1 - \cos 2\theta) = 1 - \cos 2\theta.$$

Substitute (2) and (3) in (1), we get

$$(1) \Rightarrow I = \int_0^{\pi/4} \sqrt{1 - \cos 2\theta} (-2 \sin 2\theta) d\theta = 2 \int_{\pi/4}^0 (1 - \cos 2\theta) d\theta = 2[\theta - \sin 2\theta]_{\pi/4}^0 = 2[\pi/4 - 1/2] = \pi/2 - 1$$

$$(iv) \int_{\pi/2}^0 (1 + \sin x + \cos x) dx$$

Solution:

Education

Let I

$$I = \int_{\pi/2}^0 (1 + \sin x + \cos x) dx = \int_{\pi/2}^0 1 + 2 \tan x/2 + \tan^2 x/2 + 1 - \tan^2 x/2 + \tan^2 x/2 = \int_{\pi/2}^0 1 + \tan^2 x/2 + 2 \tan x/2 + 1 - \tan^2 x/2 dx = \int_{\pi/2}^0 2 [\sec^2 x/2 + \tan x/2] dx = 2 [\tan x/2 + \log \sec(x/2)]_{\pi/2}^0$$

$$(1/2)]_{\pi/2}^0 = \tan \pi/4 + 2 \log \sec(\pi/4) - \tan 0 - 2 \log \sec 0 = 1 + 2 \log \sqrt{2} = 1 + \log 2.$$

$$(v) \int_{\pi/2}^0 \sqrt{\cos \theta} \sin^3 \theta d\theta$$

Solution:

$$I = \int_{\pi/2}^0 \sqrt{\cos \theta} \sin^2 \theta \sin \theta d\theta$$

x	0	$\frac{\pi}{2}$
t	1	0

Put $t = \cos \theta$

Differentiate with respect to ' θ '

$$dt/d\theta = -\sin \theta \Rightarrow dt = -\sin \theta d\theta$$

$$\sin^2 \theta = 1 - \cos^2 \theta = 1 - t^2$$

Substitute (2) and (3) in (1), we get,

$$(1) \Rightarrow I = \int_0^1 t(1 - t^2)(-dt) = \int_1^0 (t - t^3) dt = (t^2/2 - t^4/4)_1^0 = 1/2 - 1/4 = 1/4.$$

$$(vi) \int_0^1 \frac{1 - x^2}{(1 + x^2)^2} dx$$

Solution:

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$$\text{Let } I = \int_0^1 \frac{1 - x^2}{(1 + x^2)^2} dx = \int_0^1 [2(1 + x^2)^{-2} - 1 + x^2(1 + x^2)^{-2}] dx = \int_0^1 [2(1 + x^2)^{-2} - 1(1 + x^2)^{-1}] dx$$

$$= \int_0^1 2(1 + x^2)^{-2} dx - \int_0^1 \frac{1}{1 + x^2} dx$$

Put $x = \tan \theta \Rightarrow dx = \sec^2 \theta d\theta$

$$= 2 \int_0^{\pi/4} \frac{2 \sec^2 \theta}{(1 + \tan^2 \theta)^2} d\theta - \int_0^{\pi/4} \frac{\sec^2 \theta}{\sec^2 \theta} d\theta = 2 \int_0^{\pi/4} 2 \cos^2 \theta d\theta - \int_0^{\pi/4} 1 d\theta$$

$$= 4 \int_0^{\pi/4} \cos^2 \theta d\theta - [\theta]_0^{\pi/4} = 4 \int_0^{\pi/4} \frac{1 + \cos 2\theta}{2} d\theta - \pi/4 = 2 \int_0^{\pi/4} (1 + \cos 2\theta) d\theta - \pi/4$$

$$= 2[\theta + \sin 2\theta/2]_0^{\pi/4} - \pi/4 = 2[\pi/4 + 1/2] - \pi/4 = \pi/2 + 1 - \pi/4 = \pi/4 + 1$$

x	0	1
θ	0	$\frac{\pi}{4}$

Question 2.

$$\int 5 - 5x \cos(e^x - 1) e^x dx$$

Let $I = \int_{-5}^5 x \cos(e_x - 1e_x + 1) dx$
 $f(x) = x \cos(e_x - 1e_x + 1)$
 $f(-x) = -x \cos(1e_x - 11e_x + 1) = -x \cos(1 - e_x + e_x) = -x \cos(e_x - 11 + e_x) = -f(x)$

$$\therefore \int_{-5}^5 5x \cos(e^x - 1) dx = 0$$

Solution:

Let $I = \int_{\pi/2 - \pi/2}^{\pi/2} (x^5 + x \cos x + \tan 3x + 1) dx = \int_{\pi/2 - \pi/2}^{\pi/2} x^5 dx + \int_{\pi/2 - \pi/2}^{\pi/2} x \cos x + \int_{\pi/2 - \pi/2}^{\pi/2} \tan 3x + \int_{\pi/2 - \pi/2}^{\pi/2} dx = 0 + 0 + 0 + [x]_{\pi/2 - \pi/2}^{\pi/2} = \pi/2 - \pi/2 = 0$

(iii)

Solution:

$\therefore f(x)$ is an even function.

(iv)

Solution:

Consider:

(v) $\int_0^{2\pi} \sin^4 x \cos^3 x \, dx$

[illegible]

Solution:

$$(vii) \int \sin 2x \sin^{-1} \sqrt{t} dt + \int \cos 2x \cos^{-1} \sqrt{t} dt$$

Solution:

$$\text{Let } I = \int \sin 2x \sin^{-1} \sqrt{t} dt + \int \cos 2x \cos^{-1} \sqrt{t} dt$$

Consider:

$$I_1 = \int \sin 2x \sin^{-1} \sqrt{t} dt$$

Put $t = \sin^2 x$

Differentiate with respect to x

$$dt = 2 \sin x \cos x dx = \sin 2x dx$$

t	0	$\sin^2 x$
x	0	x

$$I_1 = \int x \sin^{-1} \sqrt{\sin^2 x} \sin 2x dx = \int x \sin^{-1}(\sin x) \sin 2x dx = \int x \sin 2x dx$$

$I_2 = \int \cos 2x \cos^{-1} \sqrt{t} dt$
Put $t = \cos^2 x$ Differentiate with respect

$$\text{to } x \quad dt = -2 \cos x \sin x dx = -\sin 2x dx$$

$$I_2 = \int_{\pi/2}^{\pi} \cos^{-1} \sqrt{\cos^2 x} (-\sin 2x) dx = -\int_{\pi/2}^{\pi} \cos^{-1}(\cos x) \sin 2x dx$$

$$I_2 = \int_{\pi/2}^{\pi} x \sin 2x dx \quad (1) \Rightarrow I = \int x \sin 2x dx + \int_{\pi/2}^{\pi} x \sin 2x dx = \int_{\pi/2}^{\pi} x \sin 2x dx$$

t	0	$\cos^2 x$
x	$\frac{\pi}{2}$	x

Apply Bernoulli's formula

$$\int u dv = [uv - u_1 v_1] = (x[-\cos 2x] -$$

$$[-\sin 2x^4])_{\pi/2}^0 = (-x \cos 2x + \sin 2x^4)_{\pi/2}^0 = -\pi^2 \cos \pi^2 + \sin \pi^4 = -\pi^2(-12) + 0 = \pi^4$$

$$(viii) \int_{10} \log(1+x)^{1+x^2} dx$$

Solution:

$$\text{Let } I = \int_{10} \log(1+x)^{1+x^2} dx \quad \text{Put } x = \tan \theta$$

Differentiate with respect to θ

x	0	1
θ	0	$\frac{\pi}{4}$

$$dx = \sec^2 \theta d\theta$$

Substitute (2) and (3) in (1), we get

$$(1) \Rightarrow I = \int_{\pi/4}^0 \log(1+\tan \theta)^{1+\tan^2 \theta} \sec^2 \theta d\theta = \int_{\pi/4}^0 \log(1+\tan \theta) \sec^2 \theta \sec^2 \theta d\theta = \int_{\pi/4}^0 \log(1+\tan \theta) d\theta$$

$$\int_0^a f(x) dx = \int_0^a f(a-x) dx$$

$$I = \int_{\pi/4}^0 \log[1+\tan(\pi/4-\theta)] d\theta = \int_{\pi/4}^0 \log[1+1-\tan \theta \frac{1+\tan \theta}{1-\tan \theta}] d\theta = \int_{\pi/4}^0 \log[1+\tan \theta + 1-\tan \theta \frac{1+\tan \theta}{1-\tan \theta}] d\theta = \int_{\pi/4}^0 \log[2] d\theta = \int_{\pi/4}^0 \log 2 d\theta = \log 2 [\theta]_{\pi/4}^0 = \pi/4 \log 2 \Rightarrow I = \pi/8 \log 2.$$

(ix)

$$\int_{\pi/10}^0 x \sin x^{1+\sin x} dx$$

Solution:

$$\text{Let } I = \int_0^\pi x \sin x \, dx$$

$$\int_a^b f(x) \, dx = \int_a^b f(a-x) \, dx \quad I = \int_0^\pi (\pi-x) \sin x \, dx$$

$$[\because \sin(\pi-x) = \sin x]$$

$$(1)+(2) \quad 2I = \int_0^\pi x \sin x \, dx + \int_0^\pi (\pi-x) \sin x \, dx \quad 2I = \int_0^\pi (x+\pi-x) \sin x \, dx$$

$$x = \pi \int_0^\pi \sin x \, dx = \pi \int_0^\pi \sin x \, dx = \pi [-\cos x]_0^\pi = \pi [-\cos \pi + \cos 0] = \pi [1+1] = 2\pi$$

$$2I = 2\pi \Rightarrow I = \pi$$

$$\int_{3\pi/8}^{11\pi/8} \sqrt{\tan x} \, dx \quad \text{Solution: Let } I = \int_{3\pi/8}^{11\pi/8} \sqrt{\tan x} \, dx$$

$$\int_a^b f(x) \, dx = \int_a^b f(a+b-x) \, dx \quad I = \int_{3\pi/8}^{11\pi/8} \sqrt{\tan(11\pi/8 - x)} \, dx$$

$$= \int_{3\pi/8}^{11\pi/8} \sqrt{\tan(\pi/2 - x)} \, dx = \int_{3\pi/8}^{11\pi/8} \sqrt{\cot x} \, dx = \int_{3\pi/8}^{11\pi/8} \frac{1}{\sqrt{\tan x}} \, dx$$

$$I = \int_{3\pi/8}^{11\pi/8} \sqrt{\tan x} \, dx + \int_{3\pi/8}^{11\pi/8} \frac{1}{\sqrt{\tan x}} \, dx$$

$$(1) + (2) \Rightarrow 2I = \int_{3\pi/8}^{11\pi/8} \left(\sqrt{\tan x} + \frac{1}{\sqrt{\tan x}} \right) \, dx$$

$$= \int_{3\pi/8}^{11\pi/8} \frac{1 + \tan x}{\sqrt{\tan x}} \, dx = \int_{3\pi/8}^{11\pi/8} \frac{1 + \tan^2 x}{\sqrt{\tan x}} \, dx = \int_{3\pi/8}^{11\pi/8} \frac{\sec^2 x}{\sqrt{\tan x}} \, dx$$

$$= \int_{3\pi/8}^{11\pi/8} \sec^2 x \cdot \frac{1}{2} \tan^{-1/2} x \, dx = \frac{1}{2} \int_{3\pi/8}^{11\pi/8} \sec^2 x \tan^{-1/2} x \, dx$$

$$= \frac{1}{2} \left[2 \tan^{1/2} x \right]_{3\pi/8}^{11\pi/8} = \left[\sqrt{\tan x} \right]_{3\pi/8}^{11\pi/8}$$

$$= \sqrt{\tan(11\pi/8)} - \sqrt{\tan(3\pi/8)} = \sqrt{1} - \sqrt{1} = 0$$

$$2I = 0 \Rightarrow I = 0$$

$$(xi) \int_0^\pi x [\sin^2(\sin x) + \cos^2(\cos x)] \, dx$$

Solution:

$$\text{Let } I = \int_0^\pi x [\sin^2(\sin x) + \cos^2(\cos x)] \, dx$$

$$\int_a^b f(x) \, dx = \int_a^b f(a-x) \, dx \quad I = \int_0^\pi (\pi-x) [\sin^2(\sin x) + \cos^2(\cos x)] \, dx$$

$$(1)+(2) \Rightarrow 2I = \pi \int_0^\pi [\sin^2(\sin x) + \cos^2(\cos x)] \, dx \therefore I = \frac{\pi}{2} \int_0^\pi [\sin^2(\sin x) + \cos^2(\cos x)] \, dx$$

$$\text{But } \int_a^b f(2a-x) \, dx = \int_a^b f(x) \, dx$$

$$\text{So, } I = \frac{\pi}{2} \times 2 \int_0^{\pi/2} [\sin^2(\sin x) + \cos^2(\cos x)] \, dx$$

$$\text{But, } I = \pi \int_0^{\pi/2} [\sin^2(\cos x) + \cos^2(\sin x)] \, dx$$

$$\therefore \int_a^b f(x) \, dx = \int_a^b f(a-x) \, dx$$

$$(3) + (4) \Rightarrow 2I = \pi(2) \int_0^{\pi/2} dx = 2\pi \left(\frac{\pi}{2} \right) = \pi^2 \Rightarrow I = \frac{\pi^2}{2}$$

Exercise 9.4-Additional Problems

Additional Problems

Question 1.

$$\text{Evaluate: } \int_0^{2\pi} \cos x \sqrt{4+3\sin x} \, dx$$

Solution:

$$I = \int_0^{2\pi} \cos x \sqrt{4+3\sin x} \, dx$$

$$\text{Put } 4+3\sin x = t, \text{ so that } 3\cos x \, dx = dt \Rightarrow \cos x \, dx = \frac{1}{3} dt$$

$$\text{When } x=0, t=4+3(0)=4 \text{ and when } x=2\pi, t=4+3(0)=4$$

$$\therefore I = \frac{1}{3} \int_4^4 \sqrt{t} \, dt = \frac{1}{3} \int_4^4 \sqrt{t} \, dt = \frac{1}{3} \left[\frac{2}{3} t^{3/2} \right]_4^4 = \frac{1}{3} \left[\frac{2}{3} (4)^{3/2} - \frac{2}{3} (4)^{3/2} \right] = \frac{1}{3} (0) = 0$$

Question 2.

$$\text{Evaluate: } \int_{\pi/4}^{\pi/2} \sin^3 x \cos^5 x \, dx$$

Solution:

$$I = \int_{\pi/4}^{\pi/2} \sin^3 x \cos^5 x \, dx = \int_{\pi/4}^{\pi/2} \sin^2 x \cos^4 x \cdot \sin x \, dx = \int_{\pi/4}^{\pi/2} (1-\cos^2 x) \cos^4 x \sin x \, dx$$

$$\text{Put } \cos x = t, \text{ so that } -\sin x \, dx = dt$$

$$\therefore I = \int_{10} t^3 dt = \left[\frac{t^4}{4} \right]_{10} = 14 - 0 = 14.$$

Evaluate: $\int_{\pi/20}^{\pi/10} \sin \theta \cos 5\theta d\theta$

$$I = \int_{\pi/20}^{\pi/2} \sqrt{\sin \theta} \cos^5 \theta d\theta = \int_{\pi/20}^{\pi/2} \sqrt{\sin \theta} \cos^4 \theta \cos \theta d\theta = \int_{\pi/20}^{\pi/2} \sqrt{\sin \theta} (1 - \sin^2 \theta)^2 \cos \theta d\theta$$
$$\therefore I = \int_{10} \sqrt{t} (1 - t_2) 2 dt = \int_{10} t_{12} (1 - 2t_2 + t_4) dt = \int_{10} (t_{12} - 2t_{52} + t_{92}) dt = \left[\frac{t_{32}^2}{2} - 2 \cdot \frac{t_{72}^2}{2} + \frac{t_{112}^2}{2} \right]_{10} = (23 - 47 + 211) - (0) = 154 - 132 + 42231 = 64231.$$
$$\therefore I = \int_{20}^{t_2} dt = [\tan^{-1} t]_{20}^{t_2} = \tan^{-1} t_2 - \tan^{-1} 20 = \tan^{-1} t_2 - \pi/4.$$
$$\therefore I = \int_{102}^{t_1+t_2} 5 + 4(2t_1+t_2) = 2 \int_{10}^{t_2} 5(1+t_2) + 8t = 25 \int_{10}^{t_2} dt t_2 + 85t + 1 = 25 \int_{10}^{t_2} dt (t_2 + 85t + 1625) + 925 = 25 \int_{10}^{t_2} dt (t + 45) + (35) = \left[\frac{25 \times 135 \tan^{-1} t + 4535}{1} \right]_{10} = 23[\tan^{-1} 15t + 43]_{10} = 23[\tan^{-1} 13 - \tan^{-1} 43] = 23 \tan^{-1} 13 - 43 \times 1 + 3(43) = 23 \tan^{-1} 1515 = 23 \tan^{-1} 113.$$

Consider, $t = \tan x$

x	0	$\frac{\pi}{2}$
t	0	∞

$$dt = \sec^2 x dx$$

Substitute (2) and (3) in (1), we get

$$(1) \Rightarrow \int_{\infty}^0 dt t^2 + 6 \left\{ \int dx a^2 + x^2 = 1 a \tan^{-1}(x a) \tan^{-1}(\infty) = \pi/2 \right\} \int_{\infty}^0 dt t^2 + (\sqrt{6})^2 = [1\sqrt{6} \tan^{-1} t \sqrt{6}]_{\infty}^0 = 1\sqrt{6} \tan^{-1}(\infty) - 0 = 1\sqrt{6}(\pi/2) = \pi/2 \sqrt{6}.$$

(ii) $\int_{\pi/2}^0 dx 5 + 4 \sin 2x$

Solution:

Let $I = \int_{\pi/2}^0 dx 5 + 4 \sin 2x$

Dividing both Numerator and Denominator by $\cos^2 x$

$$= \int_{\pi/2}^0 \frac{1 \cos^2 x 5 \cos^2 x + 4 \sin 2x \cos^2 x}{\cos^2 x} dx = \int_{\pi/2}^0 \sec^2 x 5 \sec^2 x + 4 \tan 2x dx = \int_{\pi/2}^0 \sec^2 x 5 (1 + \tan^2 x) + 4 \tan 2x dx = \int_{\pi/2}^0 \sec^2 x 5 + 9 \tan 2x dx$$

Consider, $t = \tan x$

Differentiate with respect to 'x' $dt = \sec^2 x dx$

x	0	$\frac{\pi}{2}$
t	0	∞

Substitute (2) and (3) in (1), we get

$$(1) \Rightarrow \int_{\infty}^0 dt 5 + 9 t^2 = 19 \int_{\infty}^0 dt 5 + t^2 = 19 \int_{\infty}^0 dt (\sqrt{5} 3)^2 + t^2 = 19 \left[\left[1\sqrt{5} 3 \tan^{-1} t \sqrt{5} 3 \right] \right]_{\infty}^0 = 19 [3\sqrt{5}(\pi/2) - 0] = 19 [3\pi/2 \sqrt{5}] = \pi/6 \sqrt{5}.$$

Ex 9.6

Question 1.

(i) $\int_{\pi/2}^0 \sin 10x dx$

Solution:

$$\int_{\pi/2}^0 \sin 10x dx = 910 \times 78 \times 56 \times 34 \times 12 \times \pi/2 = 63\pi 512$$

(ii)

$$\int_{\pi/2}^0 \cos 7x dx$$

Solution:

$$\int_{\pi/2}^0 \cos 7x dx = 67 \times 45 \times 23 = 1635$$

(iii) $\int_{\pi/4}^0 \sin 62x dx$

Solution

x	0	$\frac{\pi}{4}$
t	0	$\frac{\pi}{2}$

$$\int_{\pi/4}^0 \sin 62x dx \text{ Put } t = 2x$$

Differentiate with respect to 'x'

$$dt = 2 dx \Rightarrow dt/2 = dx$$

Substitute (2) and (3) in (1)

$$(1) \Rightarrow \int_{\pi/2}^0 \sin 6t dt/2 = 12 \int_{\pi/2}^0 \sin 6t dt = 12 [56 \times 34 \times 12 \times \pi/2] = 5\pi 64.$$

(iv) $\int_{\pi/6}^0 \sin 53x dx$

Solution:

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x	0	$\frac{\pi}{4}$
t	0	$\frac{\pi}{2}$

Put $t=3x=13 \int_{\pi/2}^0 \sin t \, dt = 13[45 \times 23] = 845$.

(v) $\int_{\pi/2}^0 \sin 2x \cos 4x \, dx$

Solution:

$\int_{\pi/2}^0 \sin 2x \cos 4x \, dx = \int_{\pi/2}^0 (1 - \cos 2x) \cos 4x \, dx = \int_{\pi/2}^0 (\cos 4x - \cos 6x) \, dx = \int_{\pi/2}^0 \cos 4x \, dx - \int_{\pi/2}^0 \cos 6x \, dx$
 $= (34 \times 12 \times \pi/2) - (56 \times 34 \times 12 \times \pi/2) = 3\pi/16 - 5\pi/32 = \pi/16 [3 - 5] = \pi/16 [6 - 52] = \pi/16 (12) = \pi/32$.

(vi) $\int_{2\pi}^0 \sin 7x \, dx$

Solution:

x	0	2π
t	0	$\frac{\pi}{2}$

Put $t=x/4=4 \int_{\pi/2}^0 \sin 7t \, dt = 4[67 \times 45 \times 23] = 6435$.

(vii) $\int_{\pi/2}^0 \sin 3\theta \cos 5\theta \, d\theta$

Solution:

We know that,

$\int_{\pi/2}^0 \sin m x \cos n x \, dx = [(m-1)(m-3)\dots 2.1][(n-1)(n-3)\dots 2.1](m+n)(m+n-2)(m+n-4)\dots$

Here $m=3, n=5$

$\int_{\pi/2}^0 \sin 3\theta \cos 5\theta \, d\theta = [(3-1)][(5-1)(5-3)](3+5)(3+5-2)(3+5-4)(3+5-6) = (2)(4)(2)(8)(6)(4)(2) = 124$.

(viii) $\int_{10}^0 x^2(1-x)^3 \, dx$

Solution:

We know that,

$\int_{10}^0 x^m(1-x)^n \, dx = m! \times n! / (m+n+1)! \int_{10}^0 x^2(1-x)^3 \, dx = 2! \times 3! / (2+3+1)!$
 $= 2 \times 6 / 1 \times 2 \times 3 \times 4 \times 5 \times 6 = 160$.

Ex 9.7

Question 1.

Evaluate the following

(i) $\int_{\infty}^0 x^5 e^{-3x} \, dx$

Solution:

We know that

$\int_{\infty}^0 x^n e^{-ax} \, dx = n! / a^{n+1} \therefore \int_{\infty}^0 x^5 e^{-3x} \, dx = 5! / 3^{5+1} = 5! / 3^6$ (Here $n=5, a=3$)

(ii) $\int_{\pi/2}^0 e^{-\tan x} \cos 6x \, dx$

Solution:

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$$= \int_{\pi/2}^0 e^{-\tan x} \sec^6 x dx = \int_{\pi/2}^0 e^{-\tan x} \sec^4 x \sec^2 x dx \text{ Put } t = \tan x$$

x	0	$\pi/2$
t	0	∞

Differentiate with respect to 'x'

$$dt = \sec^2 x dx$$

Substitute (2) and (3) in (1), we get,

$$(1) \Rightarrow \int_{\infty}^0 e^{-t} (\sec^2 x)^2 dt (\because \sec^2 x = 1 + \tan^2 x = 1 + t^2) = \int_{\infty}^0 e^{-t} (1 + t^2)^2 dt = \int_{\infty}^0 e^{-t} (1 + t^4 + 2t^2) dt =$$

$$\int_{\infty}^0 e^{-t} dt + \int_{\infty}^0 e^{-t} t^4 dt + 2 \int_{\infty}^0 e^{-t} t^2 dt$$

By using Gamma Integral,

$$= (-e^{-t})_{\infty}^0 + 4!(1)_{\infty}^0 + 2(2!)(1)_{\infty}^0 = -e^{-\infty} + e_0 + 4! + 2(2!) =$$

$$= 0 + 1 + 24 + 4 = 29$$

Question 2.

If $\int_{\infty}^0 e^{-\alpha x^2} x^3 dx = 32, \alpha > 0$, find α

Solution:

$$\int_{\infty}^0 e^{-\alpha x^2} x^3 dx = 32 \int_{\infty}^0 e^{-\alpha x^2} x^2 \cdot x dx = 32 \text{ Put } t = x^2$$

Differentiate with respect to 'x'

$$dt = 2x dx \Rightarrow dt/2 = x dx$$

Substitute (2) and (3) in (1), we get,

$$(1) \Rightarrow$$

$$\int_{\infty}^0 e^{-\alpha t} \cdot dt/2 = 32 \Rightarrow \frac{1}{2} \int_{\infty}^0 t e^{-\alpha t} dt = 32$$

By using Gamma integral ($n=1; a=\alpha$)

x	0	∞
t	0	∞

$$12[1/\alpha^2] = 32 \Rightarrow 1/\alpha^2 = 32/12 \Rightarrow \alpha^2 = 16/3 \Rightarrow \alpha = \pm \sqrt{16/3} \Rightarrow \alpha = \pm 4/\sqrt{3} \Rightarrow \alpha = 4\sqrt{3}/3$$

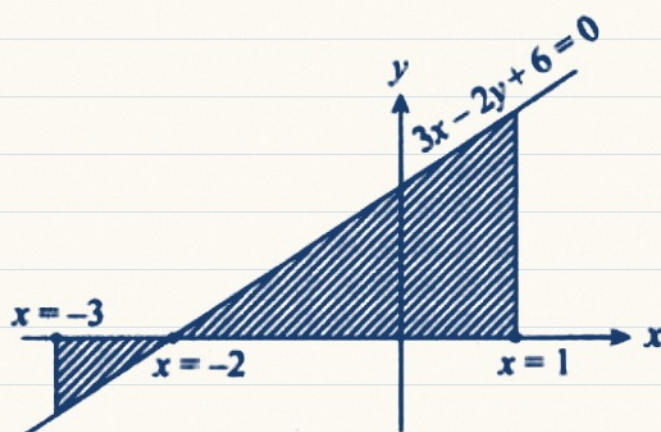
Ex 9.8

Question 1.

Find the area of the region bounded by $3x - 2y + 6 = 0, x = -3, x = 1$ and x-axis.

Solution:

$$3x - 2y + 6 = 0 \Rightarrow y = \frac{3x + 6}{2}$$



To find further limit put $y=0$

We get, $x = -2$

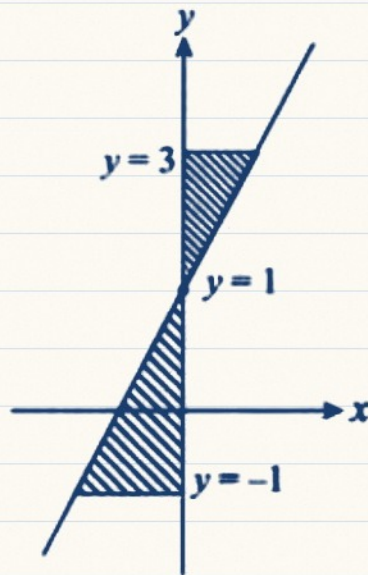
$$\begin{aligned}\text{Area} &= \int_{-2}^{-3} -y dx + \int_{-2}^{-3} y dx = -12 \int_{-2}^{-3} (3x+6) dx + 12 \int_{-2}^{-3} (3x+6) dx = -12 [3x^2 + 6x] \\ &\quad -2^{-3} + 12 [3x^2 + 6x]_{-2}^{-3} = -12 [-6 + 92] + 12 [152 + 6] = 12 [6 - 92 + 152 + 6] = 12 [12 + 3] = 152 = 7.\end{aligned}$$

5 Square units.

Question 2.

Find the area of the region bounded by $2x - y + 1 = 0$, $y = -1$, $y = 3$ and y -axis.

Solution:



$$2x - y + 1 = 0 \quad x = y - 12$$

To find further limit put $x=0$, we get $y=1$

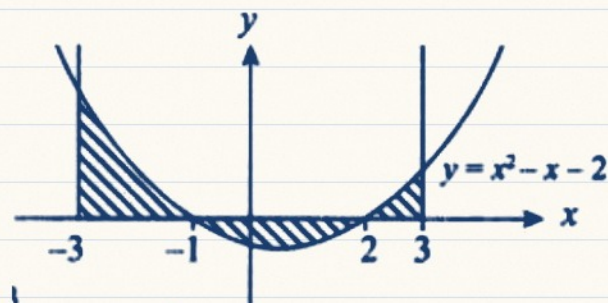
$$\begin{aligned}\text{Area} &= \int_{1-1}^{31} -x dy + \int_{31}^{92} x dy = -12 \int_{1-1}^{31} (y-1) dy + 12 \int_{31}^{92} (y-1) dy \\ &= -12 \left[\frac{y^2}{2} - y \right]_{1-1}^{31} + 12 \left[\frac{y^2}{2} - y \right]_{31}^{92} \\ &= -12 \left[\frac{(12-1)^2}{2} - (12-1) \right] + 12 \left[\frac{(92-3)^2}{2} - (92-3) \right] \\ &= -12 \left[\frac{121}{2} - 11 \right] + 12 \left[\frac{891}{2} - 89 \right] \\ &= -12 \left[\frac{121 - 22}{2} \right] + 12 \left[\frac{891 - 178}{2} \right] \\ &= -12 \left[\frac{99}{2} \right] + 12 \left[\frac{713}{2} \right] \\ &= -12 \cdot \frac{99}{2} + 12 \cdot \frac{713}{2} \\ &= -6 \cdot 99 + 6 \cdot 713 \\ &= -594 + 4278 \\ &= 3684 \text{ square units.}\end{aligned}$$

Question 3.

Find the area of the region bounded by the curve $2+x-x^2+y=0$, x -axis, $x=-3$ and $x=3$.

Solution:

To find further limits put $y=0$, we get $x=-1$ and $x=2$



Required

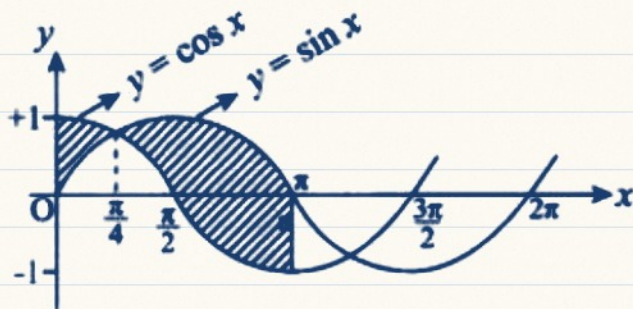
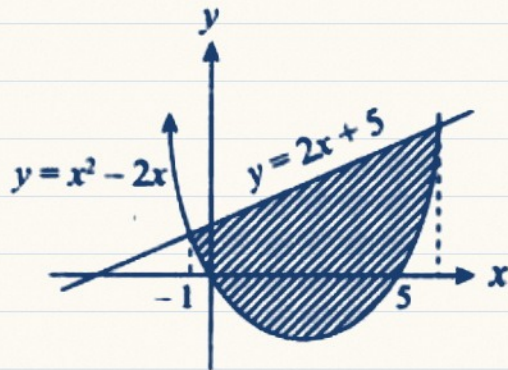
$$\begin{aligned}\text{Area} &= \int_{-1}^3 -3y \, dx + \int_{-1}^3 -y \, dx + \int_{-1}^3 2y \, dx = \int_{-1}^3 -3(x^2 - x - 2) \, dx + \int_{-1}^3 -(2 + x - x^2) \, dx + \int_{-1}^3 2(x^2 - x - 2) \, dx \\ &= (x^3 - \frac{3}{2}x^2 - 2x)_{-1}^3 + (-2x - \frac{1}{2}x^2 + \frac{1}{3}x^3)_{-1}^3 + (\frac{2}{3}x^3 - x^2 - 2x)_{-1}^3 \\ &= (27 - \frac{27}{2} - 6) - (-\frac{1}{2} + \frac{3}{2} - 2) + (-6 - \frac{9}{2} + \frac{1}{3}) - (-2 - \frac{1}{2} + \frac{1}{3}) + (2 - 1 - 2) - (-\frac{2}{3} + 1 + 2) \\ &= 526 + 276 + 116 = 906 = 15 \text{ Square units.}\end{aligned}$$

Question 4.

Find the area of the region bounded by the line $y=2x+5$ and the parabola $y=x^2-2x$.

Solution:

To find point of intersection of the curves $y=2x+5$ and $y=x^2-2x$ we get $(-1,3)$ and $(5,15)$



Area $= \int_{-1}^5 [f(x) - g(x)] dx = \int_{-1}^5 [(2x+5) - (x^2-2x)] dx = \int_{-1}^5 (4x+5-x^2) dx = [4x^2/2 + 5x - x^3/3]_{-1}^5 = (50+25-125/3) - (2-5+1/3) = 75 - 125/3 + 3 - 1/3 = 78 - 42 = 36$ Sq. units.

Question 5.

Find the area of the region bounded between the curves $y=\sin x$ and $y=\cos x$ and the lines $x=0$ and $x=\pi$.

Solution:

To find the points of intersection,

$$\sin x = \cos x = 1/\sqrt{2} \Rightarrow x = \pi/4 \quad \sin x = \cos x = -1/\sqrt{2} \Rightarrow x = 5\pi/4$$

From the diagram, we see that $\cos x > \sin x$ for $0 \leq x \leq \pi/4$ and $\sin x > \cos x$ for $\pi/4 < x \leq \pi$

$$\text{Area} = \int_0^{\pi/4} (\cos x - \sin x) dx + \int_{\pi/4}^{\pi} (\sin x - \cos x) dx = (\sin x + \cos x)_{\pi/4}^0 + (-\cos x - \sin x)_{\pi/4}^{\pi} = (1 + 1/\sqrt{2}) - (0 + 1) + (1 - 0) - (-1/\sqrt{2} - 1/\sqrt{2}) = 2\sqrt{2} \text{ sq. units.}$$

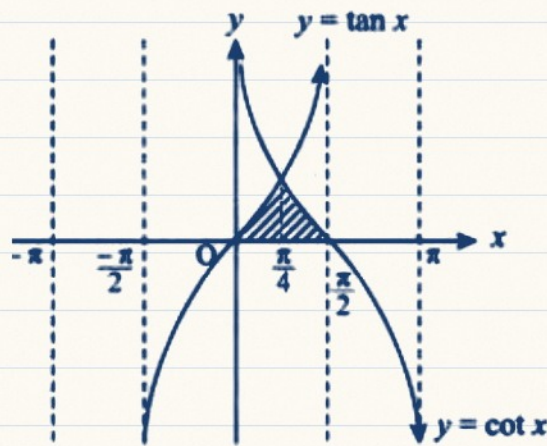
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Question 6.

Find the area of the region bounded by $y=\tan x$, $y=\cot x$ and the line $x=0, x=\pi/2, 0$

Solution:

To find the points of intersection of these two curves between 0 to $\pi/2$ is $\pi/4$



$$= \int_{\pi/4}^0 \tan x dx + \int_{\pi/2}^{\pi/4} \cot x dx = (\log \sec x)_{\pi/4}^0 +$$

$$(\log \sin x)_{\pi/2}^{\pi/4} = \log \sec(\pi/4) - \log \sec(0) + \log \sin(\pi/2) - \log \sin(\pi/4) = \log \sqrt{2} - 0 + 0 - \log 1/\sqrt{2} =$$

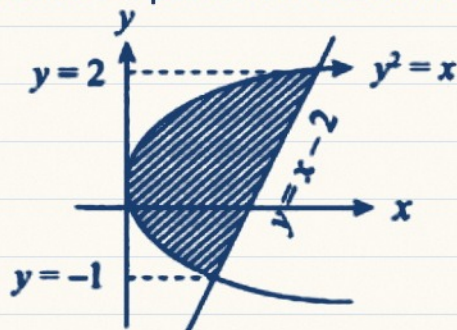
$$\log \sqrt{2} + \log \sqrt{2} = \log(\sqrt{2} \times \sqrt{2}) = \log 2$$

Question 7.

Find the area of the region bounded by parabola $y^2 = x$ and the line $y = x - 2$

Solution:

To find the points of intersection solve the two equations $y^2 = x$ and $y = x - 2$



$$\text{Area} = \int_{-1}^2 [f(y)$$

$$-g(y)] dy = \int_{-1}^2 [y+2-y^2] dy = [y^2/2 + 2y - y^3/3]_{-1}^2 = (4+4-8/3) - (1/2 - 2 + 1/3) = 9/2 \text{ Sq. units.}$$

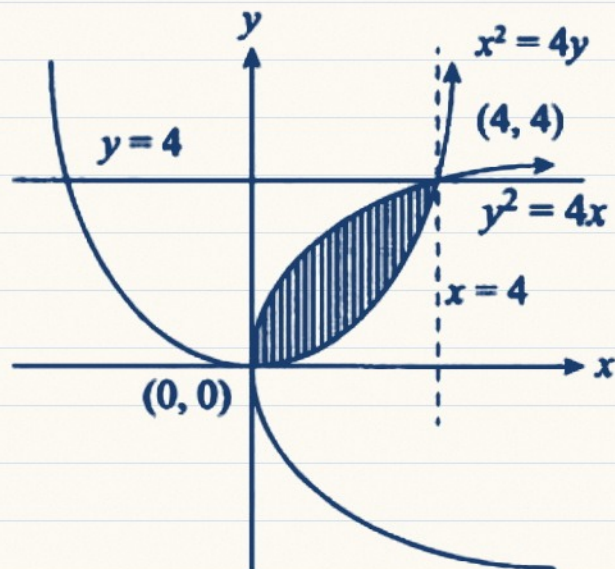
Question 8.

Father of a family wishes to divide his square field bounded by $x=0, x=4, y=4$ and $y=0$ along the curve $y^2=4x$ and $x^2=4y$ into three equal parts for his wife, daughter and son. Is it possible to divide? If so, find the area to be divided among them.

Solution:

To find the points of intersection of the two curves, $y^2=4x$ and $x^2=4y$ are $(0,0)$ and $(4,4)$.

$$\text{Area of the square field} = 4 \times 4 = 16 \text{ sq. units}$$



$\therefore$ Each of three portions must be =163 sq.units.

Now, Area of the middle portion

(Area between two curves)

$$= 43(4)^{3/2} - (4)^3 12 = 323 - 6412 = 323 - 163 = 163 \text{ sq.units.}$$

So, the remaining each of the two parts must be 163 sq.units.

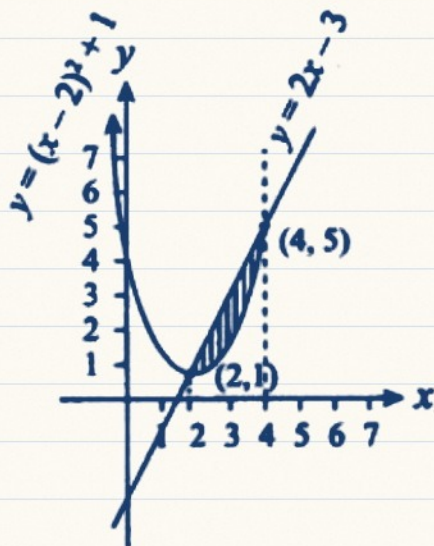
$\therefore$ Yes, It is possible to divide the square field into three equal parts.

Question 9.

The curves $y = (x-2)^2 + 1$ has a minimum point at P. A point Q on the curve is such that the slope of PQ is 2. Find the area bounded by the curve and the chord PQ.

Solution:

$$y = (x-2)^2 + 1 \quad \frac{dy}{dx} = 2(x-2)$$



$\frac{d^2y}{dx^2} = 2$ Put $\frac{dy}{dx} = 0$, we get x At $x=2$ $\frac{d^2y}{dx^2} > 0$

$\therefore x=2$ is a minimum point

$\therefore$ The point P is (2,1)

But slope of PQ is 2

$\therefore$ Equation of the chord PQ

$$y - y_1 = m(x - x_1) \quad y - 1 = 2(x - 2) \quad y - 1 = 2x - 4 \quad y = 2x - 3$$

On solving the curve and line we get the point Q(4,5)

$$\text{Required Area} = \int_2^4 (f(x) - g(x)) dx = \int_2^4 (2x - 3 -$$

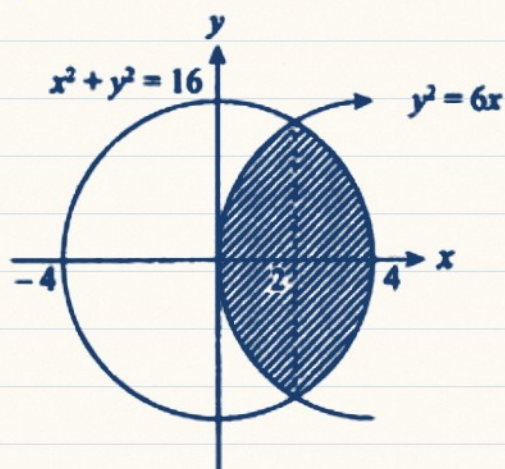
$$[(x-2)^2+1]dx = \int_2^4 (6x-x^2-8)dx = [6x^2/2 - x^3/3 - 8x]_2^4 = (48-64/3-32) - (12-8/3-16) = 20 - 56/3 = 43/3 \text{ sq. units.}$$

Question 10.

Find the area of the region common to the circle $x^2+y^2=16$ and the parabola $y^2=6x$.

Solution:

To find points of intersection of $x^2+y^2=16$ and $y^2=6x$ are $(2, 2\sqrt{3})$ and $(2, -2\sqrt{3})$. Due to symmetrical property,



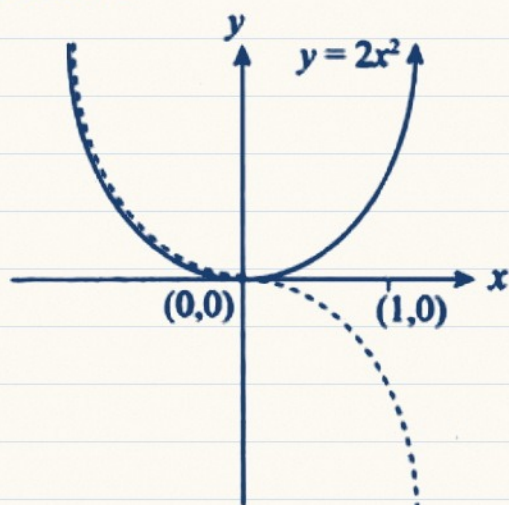
$$\begin{aligned} \text{Area} &= 2 \int_0^2 \sqrt{6x} dx + 2 \int_2^4 \sqrt{16-x^2} dx = 2\sqrt{6} \left[\frac{2}{3} x^{3/2} \right]_0^2 + 2 \left[x \sqrt{16-x^2} + 16 \sin^{-1} \left(\frac{x}{4} \right) \right]_2^4 \\ &= 8\sqrt{12} - 2\sqrt{12} + 8\pi - 8\pi/3 = 43(4\pi + \sqrt{3})/3 \text{ sq. units.} \end{aligned}$$

Ex 9.9

Question 1.

Find, by integration, the volume of the solid generated by revolving about the X-axis, the region enclosed by $y=2x^2$, $y=0$ and $x=1$

Solution:



$$V = \int_0^1 \pi y^2 dx = \int_0^1 \pi (2x^2)^2 dx = 4\pi \int_0^1 x^4 dx = 4\pi \left[\frac{x^5}{5} \right]_0^1 = \frac{4\pi}{5} \text{ cubic units.}$$

Question 2.

Find, by integration, the volume of the solid generated by revolving about the X-axis, the region enclosed by $y=e^{-2x}$, $y=0$, $x=0$ and $x=1$

Solution:

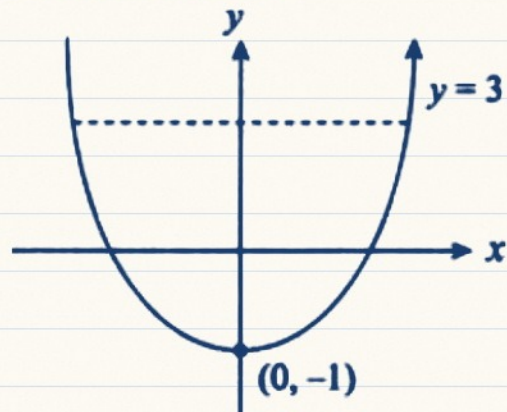
$$V = \int_0^1 \pi y^2 dx = \int_0^1 \pi (e^{-2x})^2 dx = \pi \int_0^1 e^{-4x} dx = \pi \left[-\frac{e^{-4x}}{4} \right]_0^1 = \frac{\pi}{4} [1 - e^{-4}] \text{ cubic units.}$$

Question 3.

Find, by integration, the volume of the solid generated by revolving about the y-axis, the region enclosed by $x^2 = 1 + y$ and $y = 3$.

Solution:

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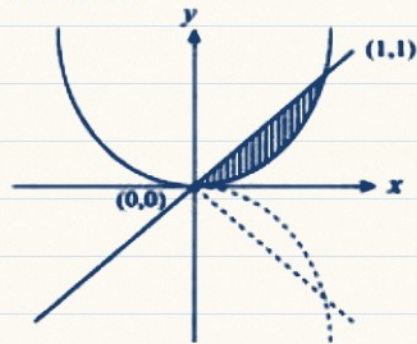


$$V = \int_{-1}^3 \pi x^2 dy = \int_{-1}^3 \pi(1+y) dy = \pi[y + y^2/2]_{-1}^3 = \pi[(3+9/2) - (-1+1/2)] = \pi[15/2 + 1/2] = \pi[16/2] = 8\pi \text{ cubic units.}$$

Question 4.

The region enclosed between the graphs of $y = x$ and $y = x^2$ is denoted by R, Find the volume generated when R is rotated through 360° about x - axis.

Solution:



To find points of intersection, solving $y = x^2$ and $y = x$, we get $(0,0)$ and $(1,1)$

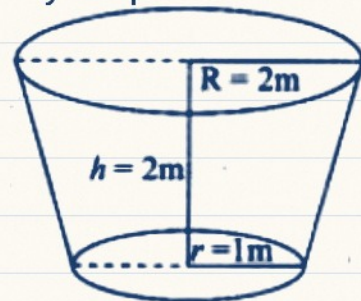
$$\text{Volume} = \pi \int_0^1 [(f(x))^2 - (g(x))^2] dx = \pi \int_0^1 [(x)^2 - (x^2)^2] dx = \pi \int_0^1 (x^2 - x^4) dx = \pi [x^3/3 - x^5/5]_0^1 = \pi [1/3 - 1/5] = 2\pi/15 \text{ cubic units.}$$

Question 5.

Find, by integration, the volume of the container which is in the shape of a right circular conical frustum as shown in the Figure.

Solution:

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$$\text{Volume of the frustum} = \int_0^h \pi [(R - r)(R - r - y)h] dy$$

$\int_0^2 \pi [(4-y)^2 - y^2] dy = \pi \int_0^2 (16 - 8y + y^2 - y^2) dy = \pi \int_0^2 (16 - 8y) dy = \pi [16y - 4y^2]_0^2 = \pi [32 - 16] = 16\pi$

Question 6.

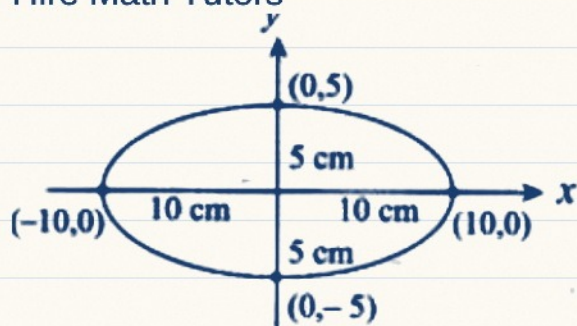
A watermelon has an ellipsoid shape which can be obtained by revolving an ellipse with major-axis 20 cm and minor-axis 10 cm about its major-axis. Find its volume using integration.

Solution:

From the given data $a=10$ cm and $b=5$ cm

Equation of the Ellipse

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$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \Rightarrow \frac{x^2}{100} + \frac{y^2}{25} = 1 \Rightarrow \frac{y^2}{25} = 1 - \frac{x^2}{100}$
 Volume = $\int_{-10}^{10} \pi y^2 dx = \pi \int_{-10}^{10} 25 \left(1 - \frac{x^2}{100}\right) dx = 25\pi \int_{-10}^{10} \left(1 - \frac{x^2}{100}\right) dx$
 $= 25\pi \left[x - \frac{x^3}{300} \right]_{-10}^{10} = 25\pi \left[\left(10 - \frac{1000}{300}\right) - \left(-10 + \frac{1000}{300}\right) \right] = 25\pi \left[\frac{2000}{300} \right] = \frac{2000\pi}{3} \text{ cm}^3$

Ex 9.10

Choose the correct or the most suitable answer from the given four alternatives:

Question 1.

The value of $\int_0^2 x \sqrt{4-9x^2} dx$ is

- (a) $\frac{\pi}{6}$
- (b) $\frac{\pi}{2}$
- (c) $\frac{\pi}{4}$
- (d) π

Solution:

- (a) $\frac{\pi}{6}$

Hint:

$\int_0^2 x \sqrt{4-9x^2} dx = \left[-\frac{1}{27} (4-9x^2)^{3/2} \right]_0^2 = -\frac{1}{27} (4-36)^{3/2} = -\frac{1}{27} (-32)^{3/2} = \frac{1}{27} (32)^{3/2} = \frac{1}{27} (32 \cdot \sqrt{32}) = \frac{1}{27} (32 \cdot 4\sqrt{2}) = \frac{128\sqrt{2}}{27}$

Question 2.

The value of $\int_{-1}^2 |x| dx$ is

- (a) 12
- (b) 32
- (c) 52
- (d) 72

Solution:

- (c) 52

Hint:

$\int_{-1}^2 |x| dx = \int_{-1}^0 -x dx + \int_0^2 x dx = \left[-\frac{x^2}{2} \right]_{-1}^0 + \left[\frac{x^2}{2} \right]_0^2 = \frac{1}{2} + 2 = \frac{5}{2}$

Question 3.

For any value of $n \in \mathbb{Z}$, $\int_0^{\pi} e^{\cos 2x} \cos 3[(2n+1)x] dx$ is

- (a) $\frac{\pi}{2}$
- (b) π

(c) 0

(d) 2

Solution:

(c) 0

Hint:

$$I = \int_{\pi/2}^{\pi/2} e^{\cos 2x} \cos 3[(2n+1)x] dx \int_a^a f(x) dx = \int_a^a f(a-x) dx \quad I = - \int_{\pi/2}^{\pi/2} e^{\cos 2x} \cos 3[(2n+1)x] dx \quad (1) + (2) \Rightarrow 2I = 0 \Rightarrow I = 0.$$

Question 4.

The value of $\int_{\pi/2}^{\pi/2} \sin 2x \cos x dx$ is.

(a) 32

(b) 12

(c) 0

(d) 23

Solution:

(d) 23

Hint:

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It is an even function

$$\int_{\pi/2}^{\pi/2} \sin 2x \cos x dx = 2 \int_{\pi/2}^{\pi/2} \sin 2x \cos x dx = 2(\sin 3x/3)_{\pi/2}^{\pi/2} = 23.$$

Question 5.

The value of $\int_{-4}^4 [\tan^{-1}(x^2x^4+1) + \tan^{-1}(x^4+1x^2)] dx$ is.

(b) 2π

(c) 3π

(d) 4π

Solution:

(d) 4π

Hint:

$$= \int_{-4}^4 \tan^{-1} \left[\frac{x^2x^4+1+x^4+1x^2}{(x^2x^4+1)(x^4+1x^2)} \right] dx \\ = \int_{-4}^4 \tan^{-1} [\infty] dx = \pi/2 [x]_{-4}^4 = \pi/2 [4+4] = 4\pi.$$

Question 6.

The value of $\int_{\pi/4}^{\pi/4} (2x^7 - 3x^5 + 7x^3 - x + 1 \cos 2x) dx$ is..

(a) 4

(b) 3

(c) 2

(d) 0

Solution:

(c) 2

Hint:

$$= \int_{\pi/4}^{\pi/4} 2x^7 \cos 2x - \int_{\pi/4}^{\pi/4} 3x^5 \cos 2x + \int_{\pi/4}^{\pi/4} 7x^3 \cos 2x - \int_{\pi/4}^{\pi/4} x \cos 2x + \int_{\pi/4}^{\pi/4} 1 \cos 2x dx \\ = 0 - 0 + 0 - 0 + 2 \int_{\pi/4}^{\pi/4} \sec 2x dx = 2[\tan x]_{\pi/4}^{\pi/4} = 2[\tan \pi/4 - \tan(0)] = 2(1) = 2.$$

Question 7.

If $f(x) = \int_0^x t \cos t dt$, then $\frac{df}{dx} =$

(a) $\cos x - x \sin x$

(b) $\sin x + x \cos x$

(c) $x \cos x$

(d) $x \sin x$

Solution:

(c) $x \cos x$

Hint:

$$= \int_{\pi/4-\pi/4}^{\pi/4-\pi/4} 2x^7 \cos 2x - \int_{\pi/4-\pi/4}^{\pi/4-\pi/4} 3x^5 \cos 2x + \int_{\pi/4-\pi/4}^{\pi/4-\pi/4} 7x^3 \cos 2x - \int_{\pi/4-\pi/4}^{\pi/4-\pi/4} x \cos 2x + \int_{\pi/4-\pi/4}^{\pi/4-\pi/4} 1 \cos 2x dx = 0 - 0 + 0 - 0 + 2 \int_{\pi/4-\pi/4}^{\pi/4-\pi/4} \sec 2x dx = 2[\tan x]_{\pi/4-\pi/4}^{\pi/4-\pi/4} = 2[\tan \pi/4 - \tan(0)] = 2(1) = 2.$$

Question 8.

The area between $y^2=4x$ and its latus rectum is

- (a) 23
- (b) 43
- (c) 83
- (d) 53

Solution:

- (c) 83

Hint:

$$y^2=4x \Rightarrow a=1$$

$$A = 2 \int_0^a y dx = 2 \int_0^1 2\sqrt{x} dx = 4 \left(\frac{x^{3/2}}{3/2} \right)_0^1 = 83$$

Question 9.

The value of $\int_{10}^0 x(1-x)^{99} dx$ is

- (a) 111000
- (b) 110100
- (c) 110010
- (d) 110001

Solution:

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- (b) 110100

$$\text{Hint: } \left[\int_0^1 x(1-x)^n dx = \frac{1}{(n+1)(n+2)} \right] \Rightarrow \int_{10}^0 x(1-x)^{99} dx = \frac{1}{100(101)} = 110100.$$

Question 10.

The value of $\int_0^\pi dx \sqrt{1+\cos x}$ is

- (a) π^2
- (b) π
- (c) $3\pi^2$
- (d) 2π

Solution:

- (a) π^2

$$\text{Hint: } I = \int_0^\pi dx \sqrt{1+\cos x} = \int_0^\pi f(x) dx = \int_0^\pi f(a-x) dx \Rightarrow 2I = \int_0^\pi dx \sqrt{1+\cos x} = \int_0^\pi \sqrt{1+\cos x} dx = \int_0^\pi \sqrt{2\cos^2(x/2)} dx = \int_0^\pi \sqrt{2} |\cos(x/2)| dx = \int_0^\pi \sqrt{2} \cos(x/2) dx = 2\sqrt{2} \sin(x/2) \Big|_0^\pi = 2\sqrt{2} \sin(\pi/2) = 2\sqrt{2}$$

Question 11.

If $\Gamma(n+2)\Gamma(n)=90$ then n is.

- (a) 10
- (b) 5
- (c) 8
- (d) 9

Solution:

- (d) 9

Hint:

$$\sqrt{n+2}\sqrt{n} = 90 \Rightarrow (n+1)\sqrt{n+1}\sqrt{n} = 90 \Rightarrow (n+1)n\sqrt{n} = 90 \Rightarrow n^2+n-90=0 \Rightarrow (n-9)$$

$$(n+10)=0 \Rightarrow n=9.$$

Question 12.

The value of $\int_{\pi/6}^0 \cos 3x dx$ is.

- (a) 23
- (b) 29
- (c) 19
- (d) 13

Solution:

(b) 29

Hint:

$$\int_{\pi/6}^0 \cos 3x dx \text{ Put } t=3x=13 \int_{\pi/20}^0 \cos 3t dt=13(23)=29.$$

x	0	$\frac{\pi}{6}$
t	0	$\frac{\pi}{2}$

Question 13.

The value of $\int_{\pi/2}^0 \sin^4 x dx$ is.

(a) $3\pi/10$

(b) $3\pi/8$

(c) $3\pi/4$

(d) $3\pi/2$

Solution:

$$3\pi/8$$

Hint:

$$=2 \int_{\pi/2}^0 \sin^4(\pi-x) dx = 2 \int_{\pi/2}^0 \sin^4 x dx = 2 \int_{\pi/2}^0 \sin^4(2a-x) dx$$

$$=2 \int_{\pi/2}^0 \sin^4 x dx = 2(3 \times 12 \times \pi/2) = 3\pi/8.$$

Question 14.

The value of $\int_{\infty}^0 e^{-3x} x^2 dx$ is

(a) 727

(b) 527

(c) 427

(d) 227

Solution:

(d) 227

Hint:

$$\therefore \int_{\infty}^0 x^n e^{-ax} dx = n! a^{n+1} \int_{\infty}^0 e^{-3x} x^2 dx = 2!(3)^{2+1} = 227.$$

Question 15.

If $\int_a^0 14+x^2 dx = \pi/8$ then a is.

(a) 4

(b) 1

(c) 3

(d) 2

Solution:

(d) 2

Hint:

$$\int_a^0 14+x^2 dx = \pi/8 [12 \tan^{-1}(x^2)]_a^0 = \pi/8 \tan^{-1}(a^2) = \pi/4 \Rightarrow a^2 = \tan(\pi/4) = 1 \Rightarrow a = 2$$

Question 16.

The volume of solid of revolution of the region bounded by $y^2 = x(a-x)$ about X-axis is

(a) πa^3

(b) $\pi a^3/4$

(c) $\pi a^3/5$

(d) $\pi a^3/6$

Solution:

(d) $\pi a^3/6$

Hint:

$$V = \pi \int_a^b y^2 dx$$
 Put $y=0$; we

get $x=0$ and $x=a=\pi$ $\int_0^\pi a_0(ax-x^2)dx=\pi[ax^2/2-x^3/3]_0^\pi=\pi[a\pi^2/2-\pi^3/3]=\pi[a\pi^6/6]=\pi a\pi^6$.

Question 17.

If $f(x) = \int_1^x e^{\sin u} du$, $x > 1$ and $\int_1^3 e^{\sin 2x} dx = 12[f(a) - f(1)]$, then one of the possible value of a is

- (a) 3
(b) 6
(c) 9
(d) 5

Solution:

- (c) 9

Hint:

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$$x = \sqrt{t}$$

x	1	3
t	1	9

$$\int 31e^{\sin x_2} x dx \text{ Put } t = x_2 \Rightarrow dt = dx \Rightarrow \int 31e^{\sin t} \sqrt{t} \cdot dt \sqrt{t} = 12[f(a)$$
$$-f(1)] \Rightarrow 12 \int_{91} e^{\sin t} dt \Rightarrow 12 \int_{91} dx [f(x)] = 12[f(a) - f(1)]f(a) - f(1) = f(a) - f(1)a = 9.$$

Question 18.

The value of $\int_{10} (\sin^{-1} x)^2 dx$ is

- (a) $\pi_{24}-1$
(b) $\pi_{24}+2$
(c) $\pi_{24}+1$
(d) $\pi_{24}-2$

Solution:

- (d) $\pi_{24}-2$

Hint:

Put $x = \sin t$

$$dx = \cos t \, dt$$

$$= \int_{\pi/2}^0 2t^2 \cos t \, dt = [t^2(\sin t) - 2t(-\cos t) + 2(-\sin t)]_{\pi/2}^0$$

x	0	1
t	0	$\frac{\pi}{2}$

Question 19.

The value of $\int_0^a (\sqrt{a^2 - x^2})^3 dx$ is.

- (a) π_{316}
(b) $3\pi_{416}$
(c) $3\pi_{28}$
(d) $3\pi_{48}$

Solution:

- (b) 3πa₄₁₆

Hint:

Put $x = a \sin \theta$

$$dx = a \cos \theta d\theta$$

$$(\sqrt{a^2 - x^2})^3 = a^3 \cos^3 \theta = \int_{\pi/2}^0 a^3 \cos^3 \theta (a \cos \theta) d\theta = a^4 \int_{\pi/2}^0 \cos^4 \theta d\theta = a^4 [34 \times 12 \times \pi^2] = 3\pi a^4 16.$$

Question 20.

If $\int_0^x f(t) dt = x + \int_1^x t f(t) dt$, then the value of $f(1)$ is.

(a) 12

(b) 2

(c) 1

(d) 34

Solution:

(a) 12

Hint:

$$\frac{d}{dx} [\int_0^x f(t) dt] = \frac{d}{dx} [x + \int_1^x t f(t) dt] \Rightarrow f(x) = 1 + \frac{d}{dx} (1) \cdot 1 f(1) - \frac{d}{dx} (x) \cdot x f(x) f(x) = 1 + 0 - x f(x) f(x) \\ x(1+x) = 1 f(x) = 11+x \Rightarrow f(1) = 12$$